

CS 250 Software Development Lifecycle

Module Seven

Final Project

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Introduction

As part of ChadaTech's pilot transition from a traditional Waterfall development model to a Scrum-Agile framework, our team was tasked with developing an application for SNHU Travel. The goal of the project was to explore how Agile principles and Scrum practices could improve software development outcomes while supporting collaboration and adaptability. Throughout the project, I assumed various Scrum roles and created artifacts associated with each phase of the software development life cycle (SDLC). This Sprint Review and Retrospective summarizes the contributions of Scrum team roles, the completion of user stories, the team's response to changing requirements, communication practices, organizational tools, and the overall effectiveness of the Scrum-Agile approach.

Applying Scrum Roles

The success of the SNHU Travel project depended on the collaboration of the Product Owner, Scrum Master, and Development Team. Each role contributed unique responsibilities that helped move the project forward. The Product Owner was responsible for defining and prioritizing user stories that delivered value to travelers. Examples included creating personalized destination recommendations, implementing budget filtering options, and developing traveler preference profiles. By prioritizing the product backlog, the Product Owner ensured that development efforts remained focused on features that provided the greatest customer value. The Scrum Master facilitated Scrum events and helped maintain focus on sprint goals. Responsibilities included organizing Sprint Planning sessions, Sprint Reviews, and Sprint Retrospectives. The Scrum Master also encouraged communication among team members and helped address obstacles that could slow development progress.

The Development Team was responsible for designing, building, testing, and delivering software increments. Team members collaborated to implement features such as vacation package filtering, recommendation functionality, traveler dashboards, and user account management. Through iterative development, the team delivered working software while continuously incorporating feedback and improvements (Atlassian, 2024; Talreja, 2025).

By working together, these roles promoted accountability, transparency, and collaboration, which contributed significantly to project success. The Scrum-Agile approach played a critical role in helping user stories move from concept to completion.

Completing User Stories

User stories provided a customer-focused method for capturing requirements and ensuring development efforts aligned with user needs. One example user story from the project stated: "As a traveler, I want to filter vacation packages by budget so that I can quickly find trips within my price range."

The story began in the Product Backlog, where it was prioritized by the Product Owner. During Sprint Planning, the story was selected for implementation and moved into the Sprint Backlog. The Development Team then worked on building and testing the functionality. During Sprint Reviews, stakeholders evaluated the feature and provided feedback. Once the feature met acceptance criteria, it was approved and released. This iterative process allowed the team to deliver value incrementally rather than waiting until the end of development. Frequent reviews ensured that the final feature met stakeholder expectations and user requirements (Talreja, 2025).

Handling Changing Requirements

One of the greatest advantages of the Scrum-Agile approach was its ability to accommodate changing requirements. Throughout the SNHU Travel project, requirements evolved as stakeholders refined their vision and provided feedback regarding desired functionality. For example, travel-related features such as destination recommendations, traveler preferences, and filtering capabilities required ongoing refinement as the project progressed. Rather than disrupting the entire project, the Scrum framework allowed changes to be added to the Product Backlog and prioritized for future sprints. The team could adapt quickly while maintaining progress toward sprint goals.

If a traditional Waterfall approach had been used, changes would have required revisiting earlier phases of development, increasing both cost and project duration. Agile's flexibility allowed the team to respond efficiently to new information and changing customer needs while continuing to deliver working software increments (Day, 2025; GeeksforGeeks, 2026).

Communication

Effective communication was essential throughout the project. Scrum events provided structured opportunities for collaboration and information sharing. Sprint Planning sessions allowed team members to discuss priorities and determine which user stories would be included in the sprint. These discussions ensured that everyone understood project goals and expected outcomes. Sprint Reviews provided opportunities to demonstrate completed work and gather stakeholder feedback. These reviews helped ensure that development remained aligned with customer expectations and business objectives. Sprint Retrospectives encouraged reflection and continuous improvement. Team members evaluated what worked well, identified challenges, and discussed opportunities to improve future sprints (Talreja, 2025).

Throughout the course, team discussions and collaboration activities provided opportunities to practice effective communication. During Agile planning discussions, team members supported user stories with clearly defined acceptance criteria and emphasized the importance of continuous testing. In another discussion, teammates collaborated to reach consensus on Agile practices that would benefit the project. These communications encouraged participation, promoted transparency, and helped the team develop a shared understanding of project goals. As a result, collaboration improved and team members remained aligned on project priorities.

Organizational Tools and Scrum Principles

Several organizational tools and Scrum principles contributed to the success of the SNHU Travel project. The Product Backlog served as a centralized list of user stories and project requirements. It helped the Product Owner prioritize work and ensure that development focused on the most important user needs. The Sprint Backlog identified the work selected for a specific sprint and provided clear objectives for the Development Team. The Scrum Board offered visibility into project progress by tracking work through stages such as To Do, In Progress, Review, and Done. This tool improved transparency and helped team members understand project status at any given time. User stories ensured that requirements remained focused on customer needs rather than technical specifications alone (Atlassian, 2024).

Scrum events further enhanced the effectiveness of these tools. Sprint Planning aligned development efforts with project goals, Daily Scrums promoted accountability, Sprint Reviews facilitated stakeholder involvement, and Sprint Retrospectives supported continuous improvement. Together, these tools and events created an organized and collaborative development environment that improved productivity and project outcomes (Talreja, 2025).

Evaluating the Scrum-Agile Process

The Scrum-Agile methodology provided several benefits during the SNHU Travel project. First, it offered flexibility. Requirements could evolve throughout development without requiring major project disruptions. Second, Agile encouraged continuous stakeholder feedback. This ensured that features aligned with customer expectations and provided meaningful value. Third, incremental delivery allowed the team to release working functionality throughout the project rather than waiting until development was complete. Finally, Scrum improved transparency and collaboration through regular communication and clearly defined roles (Day, 2025; Talreja, 2025).

Challenges of Scrum-Agile

Although Agile provided many advantages, it also presented some challenges. The methodology requires frequent communication and active stakeholder participation. Without regular feedback and engagement, the benefits of Agile can be reduced. Additionally, evolving requirements can make long-term planning more difficult. Teams must remain flexible and prepared to adjust priorities throughout development. Agile also requires time for Scrum events such as Sprint Planning, Reviews, and Retrospectives. While these meetings provide value, they require ongoing commitment from team members (Day, 2025).

Was Scrum-Agile the Best Approach?

The Scrum-Agile approach was the best methodology for the SNHU Travel project. The project involved changing requirements, evolving user stories, and ongoing stakeholder feedback. These conditions aligned closely with the strengths of Agile development. The iterative nature of

Scrum allowed the team to refine features, incorporate feedback, and deliver value incrementally. Rather than treating change as a disruption, Agile embraced change as an opportunity to improve the product.

Based on the success of the SNHU Travel pilot project, I would recommend that ChadaTech continue expanding the Scrum-Agile approach across additional development teams. The methodology improved flexibility, collaboration, transparency, and responsiveness, resulting in a more effective development process.

Conclusion

The SNHU Travel project demonstrated the value of adopting a Scrum-Agile approach to software development. Scrum roles supported accountability and collaboration, user stories guided development efforts, and organizational tools improved project visibility. Agile practices enabled the team to respond effectively to changing requirements while continuously delivering customer value. Although the methodology requires ongoing communication and stakeholder involvement, the benefits significantly outweighed the challenges. Based on the results of this pilot project, Scrum-Agile proved to be the most effective approach for developing the SNHU Travel application and should be considered for broader adoption throughout ChadaTech.

References

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